

A UNIFIED DEEP-LEARNING FRAMEWORK FOR SCHWARZSCHILD QUASI-NORMAL-MODE EXTRACTION

Executive Summary

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Motivation & Scientific Justification

The ringdown of a perturbed black hole is a superposition of damped sinusoids — quasi-normal modes (QNMs) — whose frequencies and damping times are fixed entirely by the black hole’s mass and spin [1]. Measuring them from gravitational-wave data and comparing against the predictions of general relativity is among the cleanest strong-field tests available, and has been an observational programme since the ringdown analysis of GW150914 [2]. The precision of such tests rests on two computational ingredients: the accuracy of the underlying waveform solution, and the reliability of reading the QNM parameters off it. The second receives surprisingly little scrutiny — the damped-sinusoid template is valid only inside a finite time window, and the choice of window biases the recovered parameters in ways most analyses do not quantify.

Machine-learning solvers promise to relieve the first bottleneck. Physics-informed neural networks (PINNs) solve differential equations by minimising the equation residual directly; the only prior PINN treatment of the time-domain Schwarzschild problem, however, reports a 28% relative waveform error that dominates its QNM error budget, with a fit window fixed by hand [5]. This project benchmarks what PINNs and neural surrogates can deliver for QNM extraction — on the Schwarzschild $\ell = 2$ master equations, where exact reference values are known from Leaver’s continued-fraction method [3] — and builds an extraction protocol that quantifies its own fit-window systematic.

Methodology

Four models are evaluated under one fixed protocol. (i) A *reproduction* PINN, architecture-matched to the prior study [5], establishes a trustworthy baseline. (ii) An *enhanced forward* PINN doubles the quasi-Newton optimisation budget and adds residual-greedy sampling, which concentrates training points where the equation residual is currently largest. (iii) An *inverse* PINN embeds a ringdown template with five learnable scalars and recovers the black-hole mass jointly with the waveform from 500 observations at 1% noise. (iv) A *hybrid surrogate* runs a coarse finite-difference solve at 1/64 of the reference cost and trains a small Fourier neural operator [6] (1.1 million parameters) to predict its discretisation error, following the solver-correction strand

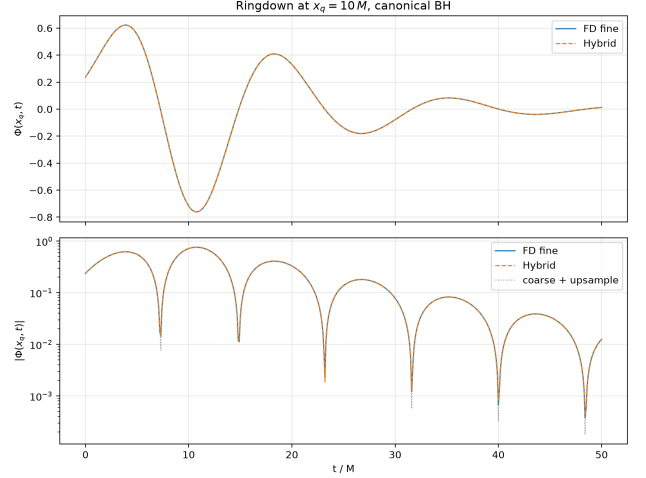


Figure 1: Ringdown waveform at the observer for the benchmark configuration, on linear (top) and logarithmic (bottom) scales: the hybrid surrogate (dashed) overlaid on the fine finite-difference reference (solid) and the 64 $\times$ -cheaper coarse baseline (dotted). The QNM parameters are extracted from such time series.

of the machine-learning literature [7, 8]; the training target is built label-free by Richardson extrapolation of two coarse solves, and a parameter-free gate confines the correction to the under-resolved wavefronts.

Extraction is deliberately model-agnostic: every model produces the same observer time series (Figure 1), from which five estimators recover the frequency and damping time (ω, τ). The methodological contribution is a *plateau scan* (Figure 2): the fit window is swept systematically (one edge or both), and the contiguous block of windows over which the recovered values are stationary supplies the estimate, its window systematic σ_ω , and a validity flag — if no plateau exists, the estimate is rejected rather than silently biased.

Key Findings

The reproduction reduces the prior study’s waveform error by an order of magnitude on both of its benchmark potentials (relative L^2 error 1.30% versus 28.06%, and 2.67% versus 23.59%). The enhanced forward PINN reaches 0.58%, from which the plateau estimator recovers the fundamental frequency to 0.029% and the damping time to 2.65%; an error budget shows the frequency measurement is limited by the window sys-

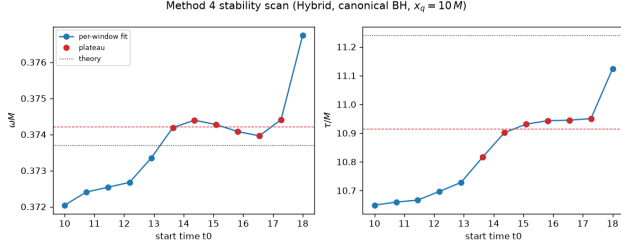


Figure 2: The plateau scan underlying the extraction protocol: recovered frequency (left) and damping time (right) as the fit-window start time is swept, for the hybrid surrogate’s benchmark waveform. Red markers form the contiguous plateau block that supplies the estimate and its window systematic; the dotted line marks the exact continued-fraction reference value.

tematic, not by the field solution. The inverse PINN recovers the black-hole mass to 0.34% under 1% observation noise — after isolating, analytically and empirically, an amplitude–damping-time degeneracy of the embedded template that would otherwise bias the damping time by ten to twenty percent. The hybrid surrogate cuts the mean-squared waveform error by a factor of 82 relative to its coarse input at 1/64 of the reference cost, with no fine simulation anywhere in its training loss; at the common observer it tracks the expensive reference’s frequency to 0.15% across a 100-black-hole population, its damping-time error (0.53%) set by the network’s late-ring residual rather than the plateau estimator.

Recommendation & Next Steps

Three follow-ups are scoped. First, the lift from Schwarzschild to rotating (Kerr) black holes — the astrophysically relevant case — for which each component here (solver, extraction protocol, hybrid design) was chosen to carry over. Second, wrapping the deterministic inverse estimate in a Bayesian posterior, so that data noise and network systematics enter a single uncertainty budget, with a Cramér–Rao bound establishing how close the 0.34% mass estimate sits to the information-theoretic limit. Third, learning the inverse map directly — from observed waveform to source parameters — across the hybrid’s training population, with the plateau diagnostic carrying over as a population-level quality flag.

Research Impact & Conclusion

The plateau estimator and its validity flag apply to any time-domain ringdown analysis, machine-learning-based or not: they replace a hand-tuned window choice with a reproducible procedure that reports its own systematic uncertainty, directly relevant as next-generation detectors push ringdown spectroscopy towards per-mille precision [4]. The hybrid result carries a broader design lesson for scientific computing: keeping a cheap classical solver responsible for the physics and learning only its discretisation error delivered reference-quality waveforms at a fraction of the cost. Every result, figure and

table in the report regenerates from the accompanying code repository.

References

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